Paper 2 · Answers & Solutions

SET-A55 · 45 questions

Part 1 · Answer Key

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
A	D	B	D	B	B	A	A	A	C	D	D	D	B	C
16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
C	D	A	C	C	B	C	C	C	D	B	C	A	D	D
31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
D	C	A	D	A	B	C	B	A	A	A	B	D	B	B

Part 2 · Worked Solutions

Section A

1 **A** — Expanding and Simplifying a Sum of Two Squares of Linear Binomials

1. $(2x+3)^2 = 4x^2 + 12x + 9$ and $(x-4)^2 = x^2 - 8x + 16$
2. Adding and combining like terms gives $5x^2 + 4x + 25$

2 **D** — The Maximum Absolute Error from a Relative Error

1. Relative error $= \dfrac{\text{maximum absolute error}}{\text{measured value}}$, so the maximum absolute error $= 48 \times \dfrac{1}{96}$
2. $= 0.5$ cm

3 **B** — The Image after a Rotation of 180 Degrees about a Fixed Point

1. The centre of rotation is the mid-point of the point and its image, so the image $= (2 \times 3 - 7,\ 2 \times 1 - (-2))$
2. $= (-1, 4)$

4 **D** — Inverse Variation: Finding a Value from a Given Pair

1. Inverse variation gives $xy = k$ constant: $k = 6 \times 15 = 90$
2. When $x = 10$, $y = \dfrac{90}{10} = 9$

5 **B** — The Third Angle at a Vertex Shared by a Regular Pentagon and a Regular Hexagon

1. An interior angle of the regular pentagon $= \dfrac{(5-2)180^\circ}{5} = 108^\circ$ and that of the regular hexagon $= \dfrac{(6-2)180^\circ}{6} = 120^\circ$
2. BA, BC and BP divide the angle at B into three angles, so $\angle CBP = 360^\circ - 108^\circ - 120^\circ = 132^\circ$

6 **B** — The Perimeter of a Rhombus from Its Diagonals

1. The diagonals bisect each other at right angles, so a side $= \sqrt{8^2 + 6^2} = 10$ cm
2. The perimeter $= 4 \times 10 = 40$ cm

7 **A** — Finding a Constant from the Solution of a System of Linear Equations

1. From $x - y = 1$, $x = y + 1$; substituting into $2x + 3y = 12$ gives $5y + 2 = 12$, so $y = 2$ and $x = 3$
2. Substituting into $kx + y = 11$ gives $3k + 2 = 11$, so $k = 3$

8 **A** — Expressing a Quotient of Powers with Different Bases as a Single Power

1. $8^{n+1} = (2^3)^{n+1} = 2^{3n+3}$ and $4^n = (2^2)^n = 2^{2n}$
2. $\dfrac{2^{3n+3}}{2^{2n}} = 2^{3n+3-2n} = 2^{n+3}$

9 **A** — Converting a Decimal Number into a Binary Number

1. $205 = 128 + 64 + 8 + 4 + 1$
2. So $205 = (11001101)_2$

10 **C** — Finding an Unknown from the Angles at the Centre Subtended by Two Equal Chords

1. Equal chords of a circle subtend equal angles at the centre, so $5x^\circ = (2x + 42)^\circ$
2. $3x = 42$, so $x = 14$

11 **D** — The Interest Rate from Simple Interest, Principal and Time

1. Simple interest: interest = principal × rate × time, so $24000 \times r \times 3 = 2880$
2. $r = \dfrac{2880}{72000} = 0.04 = 4\%$

12 **D** — The Square of the Difference of the Roots from Their Sum and Product

1. $\alpha + \beta = 7$ and $\alpha\beta = 10$
2. $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta = 49 - 40 = 9$

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13 **D** — The Shaded Area of a Square with Its Inscribed Circle Removed

1. The diameter of the inscribed circle $= 12$ cm, so the radius $= 6$ cm
2. The shaded area $= 12^2 - \pi(6)^2 = (144 - 36\pi)$ cm^2

14 **B** — Simplifying a Sum of Two Fractions and Factorizing the Numerator

1. $\dfrac{3}{x-2} + \dfrac{2}{x+3} = \dfrac{3(x+3)+2(x-2)}{(x-2)(x+3)} = \dfrac{5x}{(x-2)}$
2. Taking out the common factor 5 : $\dfrac{5(x+1)}{(x-2)(x+3)}$

15 **C** — Finding an Unknown Datum from the Median of Seven Data

1. The other six data in ascending order are 4 , 6 , 6 , 9 , 11 , 13 ; the 3rd and 4th are 6 and 9
2. So the median must be x itself, i.e. $x = 8$

16 **C** — The nth Term from the Sum of the First n Terms and the First Term

1. $S_{20} = \dfrac{20}{2}(a+l)$, so $650 = 10(5+l)$
2. $5 + l = 65$, so the 20th term $l = 60$

17 **D** — Using the Sine Formula to Find an Angle from Two Sides and a Non-included Angle

1. Sine formula: $\dfrac{\sin\angle ACB}{AB} = \dfrac{\sin\angle BAC}{BC}$, so $\sin\angle ACB = \dfrac{11\sin 40°}{9}$
2. $\sin\angle ACB = 0.78563$; taking the acute angle, $\angle ACB = 51.8°$

18 **A** — The Greatest Distance from the Fuel Consumption Rate and the Fuel Available

1. Each litre allows $\dfrac{100}{7.5}$ km
2. The distance $= 48 \times \dfrac{100}{7.5} = 640$ km

19 **C** — The Slant Height of a Right Pyramid from Its Volume and Base Edge

1. $\dfrac{1}{3}(12)^2 h = 384$, so the height $h = 8$ cm
2. The slant height $= \sqrt{8^2 + 6^2} = 10$ cm

20 **C** — The Imaginary Part of the Square of a Complex Number

1. $(3 - 4i)^2 = 9 - 24i + 16i^2 = 9 - 24i - 16$
2. $= -7 - 24i$, so the imaginary part is -24

21 **B** — Three Statements on Two Straight Lines Given in General Form

1. I is correct: $L_1 : y = 2x + 6$ and $L_2 : y = 2x - \dfrac{5}{2}$ have the same slope 2 but different y-intercepts, so they are parallel. II is correct: the y-intercept of L_1 is 6
2. III is incorrect: $4(2) - 2(2) - 5 = -1 \neq 0$, so $(2, 2)$ is not on L_2 . Hence only I and II are correct

22 **C** — Using the Method of Completing the Square to Find the Maximum Value

1. $y = -2\left(x^2 - 6x\right) - 5 = -2\left[(x-3)^2 - 9\right] - 5$
2. $= -2(x-3)^2 + 13$, so the maximum value is 13

23 **C** — The Slant Side and the Perimeter of a Right-angled Trapezium

1. Drawing the perpendicular from D to BC , the segment to $C = 14 - 8 = 6$ cm, so $CD = \sqrt{6^2 + 8^2} = 10$ cm
2. The perimeter $= 8 + 8 + 14 + 10 = 40$ cm

24 **C** — Three Statements on the Discriminant and the Sum and Product of the Roots

1. I is correct: $\Delta = (-5)^2 - 4(2)(-3) = 49 > 0$, so there are two distinct real roots. II is incorrect: $\alpha + \beta = \dfrac{5}{2}$, not $-\dfrac{5}{2}$
2. III is correct: $\alpha\beta = \dfrac{-3}{2} = -\dfrac{3}{2}$. Hence only I and III are correct

25 D — Finding a Number of Balls from a Given Probability

1. Let there be w white balls. Then $\frac{8}{8+w} = \frac{2}{5}$
2. $40 = 16 + 2w$, so $w = 12$

26 B — Three Statements on a Marked Price, a Selling Price and a Profit Percent

1. The marked price $= 500(1 + 40\%) = 700$ dollars, so I is incorrect. The selling price $= 700 \times 85\% = 595$ dollars, so II is correct
2. The profit percent $= \frac{595 - 500}{500} = 19\%$, not 25%, so III is incorrect. Hence only II is correct

27 C — The Percentage of a Class Interval Read from a Frequency Histogram

1. From the histogram the six frequencies are 3, 5, 8, 12, 7, 5, so the class has 40 students
2. The required percentage $= \frac{12}{40} \times 100\% = 30\%$

28 A — Three Statements on the Trigonometric Ratios of an Acute Angle

1. $\sin\theta = \frac{5}{13}$ with θ acute; from the triple 5, 12, 13, $\cos\theta = \frac{12}{13}$ and $\tan\theta = \frac{5}{12}$, so I and II are both incorrect
2. III is correct: $\sin\theta\cos\theta = \frac{5}{13} \times \frac{12}{13} = \frac{60}{169}$. Hence only III is correct

29 D — Three Statements on a Cubic Polynomial by the Remainder and Factor Theorems

1. $f(3) = 54 - 27 - 33 + 6 = 0$, so I is correct; $f(-2) = -16 - 12 + 22 + 6 = 0$, so $x + 2$ is a factor and II is correct
2. $(x - 3)(x + 2) = x^2 - x - 6$ and $(x^2 - x - 6)(2x - 1) = 2x^3 - 3x^2 - 11x + 6$, so III is correct. Hence I, II and III are all correct

30 D — Three Statements on a Standard Score and the Standard Deviation after Transformations

1. The standard score of Ming $= \frac{78 - 66}{8} = 1.5$, not 1.2, so I is incorrect
2. II is correct: adding the same constant to every datum leaves the standard deviation unchanged. III is correct: multiplying every datum by 1.1 multiplies the standard deviation by 1.1, i.e. $8 \times 1.1 = 8.8$ marks. Hence only II and III are correct

Section B

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31 **D** — The Distance between the Two x-intercepts of a Circle

1. Putting $y = 0$ gives $x^2 - 8x = 0$, i.e. $x(x - 8) = 0$
2. $x = 0$ or $x = 8$, so $A = (8, 0)$ and $OA = 8$

32 **C** — A Quadratic Function from Its x-intercepts and y-intercept

1. Let $y = a(x + 1)(x - 4)$. When $x = 0$, $y = -12$, so $a(1)(-4) = -12$, i.e. $a = 3$
2. $y = 3(x + 1)(x - 4) = 3x^2 - 9x - 12$

33 **A** — Three Statements on a Cyclic Quadrilateral Given One of Its Angles

1. Opposite angles of a cyclic quadrilateral are supplementary, so $\angle BCD = 180° - 104° = 76°$ and I is correct. $\angle DCE$ is an exterior angle equal to the interior opposite angle $\angle BAD = 104°$, so II is incorrect
2. $\angle BAD$ stands on the arc containing C, so the reflex $\angle BOD = 2(104°) = 208°$ and the non-reflex $\angle BOD = 360° - 208° = 152°$; III is incorrect. Hence only I is correct

34 **D** — The Third Term from the Common Ratio and the Sum to Infinity

1. $S_\infty = \dfrac{a}{1 - r}$, so $45 = \dfrac{a}{1 - \frac{2}{5}}$, i.e. $a = 45 \times \dfrac{3}{5} = 27$
2. The 3rd term $= 27\left(\dfrac{2}{5}\right)^2 = 27 \times \dfrac{4}{25} = 4.32$

35 **A** — Converting Polar Coordinates into Rectangular Coordinates

1. $x = 8\cos 150° = 8\left(-\dfrac{\sqrt{3}}{2}\right) = -4\sqrt{3}$
2. $y = 8\sin 150° = 8\left(\dfrac{1}{2}\right) = 4$, so the rectangular coordinates are $\left(-4\sqrt{3}, 4\right)$

36 **B** — The Area of a Triangle Formed by Two Lines and the Origin

1. $2x = 9 - x$ gives $x = 3$ and $y = 6$, so $P = (3, 6)$; putting $y = 0$ gives $Q = (9, 0)$
2. Taking OQ as the base, the area $= \dfrac{1}{2}(9)(6) = 27$ square units

37 **C** — The Probability that at Least One of Two Independent Events Occurs

1. The probability that neither breaks down $= (1 - 0.2)(1 - 0.1) = 0.8 \times 0.9 = 0.72$
2. The probability that at least one breaks down $= 1 - 0.72 = 0.28$

38 **B** — The Greatest Total Profit under Two Working-hour Constraints

1. Let x and y be the daily numbers of type A and type B chairs. Then $2x + y \le 18$, $x + 2y \le 21$, $x \ge 0$ and $y \ge 0$; the total profit $= 120x + 150y$
2. Solving the two constraints as equalities gives $x = 5$ and $y = 8$ with total profit $= 120(5) + 150(8) = 1800$ dollars, which exceeds the values 1080 and 1575 dollars at the other vertices

39 **A** — The Height of a Cone Recast from a Solid Hemisphere

1. The volume of the hemisphere $= \dfrac{2}{3}\pi(6)^3 = 144\pi$ cm^3
2. $\dfrac{1}{3}\pi(4)^2 h = 144\pi$, so $h = 27$ cm

40 **A** — Solving a Linear Equation with a Complex Coefficient

1. $z = \dfrac{5 + 10i}{2 + i} = \dfrac{(5 + 10i)(2 - i)}{(2 + i)(2 - i)} = \dfrac{10 - 5i + 20i}{5}$
2. $= \dfrac{20 + 15i}{5} = 4 + 3i$

41 **A** — The Largest of Three Numbers in Arithmetic Sequence Given Their Sum and Product

1. Let the numbers be $a-d$, a and $a+d$. Then $3a=42$, so $a=14$
2. $14\left(196-d^2\right)=2688$ gives $d^2=4$ and $d=2$; the numbers are 12, 14, 16 and the largest is 16

42 **B** — Three Statements on the Descriptions of Three Loci

1. I is correct: the locus of a point equidistant from a fixed point is a circle centred at that point. II is incorrect: the locus consists of two straight lines parallel to L, one on each side of L, not one
2. III is correct: the locus of a point inside the angle and equidistant from its two sides is the angle bisector. Hence only I and III are correct

43 **D** — Solving an Exponential Equation by Expressing Both Sides with the Same Base

1. $4^{x+1}=2^{2x+2}$ and $8^{x-1}=2^{3x-3}$
2. Equal powers with the same base give $2x+2=3x-3$, i.e. $x=5$

44 **B** — The Angle between an Edge and the Base of a Regular Tetrahedron

1. The height of the equilateral base $=\frac{\sqrt{3}}{2}(6)=3\sqrt{3}$ cm and the distance from the centroid to a vertex $=\frac{2}{3}\left(3\sqrt{3}\right)=2\sqrt{3}$ cm
2. $\cos\theta=\frac{2\sqrt{3}}{6}=\frac{1}{\sqrt{3}}$, so $\theta=54.7°$

45 **B** — Three Statements on Common Logarithms Evaluated from Two Given Values

1. I is correct: $\log 6=\log 2+\log 3=0.301+0.477=0.778$. II is correct: $\log 12=2\log 2+\log 3=0.602+0.477=1.079$
2. III is incorrect: $\log\frac{3}{2}=\log 3-\log 2=0.477-0.301=0.176$, while 1.585 is $\frac{\log 3}{\log 2}$. Hence only I and II are correct